

env.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/coverage/circ_bfr/env.vr

```
#include <rvm_std_lib.vrh>
#include "circ_bfr.vrh"
#include "transactions.vrh"

class test_cfg {
    rand integer run_for;
    constraint test_cfg_valid {
        run_for > 0;
    }
    constraint reasonable {
        run_for < 1000;
    }
}

class cb_env extends rvm_env {
    test_cfg cfg;
    transaction_atomic_gen gen;
    circular_buffer dut;
    task new() {
        this.cfg = new;
    }

    virtual task gen_cfg() {
        super.gen_cfg();
        void = this.cfg.randomize();
    }
    virtual task build() {
        super.build();
        this.gen = new("Source", 0);
        this.gen.stop_after_n_insts = this.cfg.run_for;
        this.dut = new;
    }
    virtual task start_t() {
        super.start_t();

        this.gen.start_xactor();

        // Use a local thread in lieu of a transactor to drive the DUT
    }
}
```

```

fork
{
    while (1) {
        transaction tr = this.gen.out_chan.get_t();
        case (tr.kind) {
            transaction::READ: {
                rvm_trace(this.log, "Reading from buffer...");
                void = this.dut.read();
            }
            transaction::WRITE: {
                rvm_trace(this.log, "Writing to buffer...");
                this.dut.write(tr.v);
            }
        }
    }
}
join none
}

virtual task wait_for_end_t() {
    super.wait_for_end_t();

    rvm_note(this.log, psprintf("Running for %0d transactions...",
                                this.cfg.run_for));

    void = this.gen.notify.wait_for_t(this.gen.DONE);
    delay(10);
}
}

```